PAUL-JASPER SAHR

SOLUTIONS ARCHITECT, FULL-STACK ENGINEER AND DESIGN THINKER

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PROFESSIONAL SUMMARY

Senior Solutions Architect & Full-Stack Engineer shipping production-grade industrial software on AWS. Architecting scalable cloud applications, GenAI solutions and ML infrastructure with TypeScript and Python. Leading technical initiatives at a major automotive joint venture, taking projects from concept to deployment while mentoring teams. Active in the local founder scene and building independent products while exploring future venture ideas. Certified Scrum Master and Design Thinking Practitioner.

PROFESSIONAL EXPERIENCE

Senior Solutions Architect & Full-Stack Engineer — since 01/2025
inpro — Berlin, Germany

- **ML Monitoring Infrastructure for Logistics ETA Models :**
  - Implemented custom data capture for SageMaker Serverless Inference endpoints
  - Set up infrastructure for monitoring dashboards and operational alerts

  Tech: AWS, SageMaker Model Monitor, CDK, CloudFormation, Athena, DynamoDB, S3, CloudWatch, Python
- **Rule Mining :**
  - Led development of a web app to extract technical rules from unstructured documents
  - Made key technical decisions, designed the system architecture, conducted code reviews and mentored a team of 2 developers
  - Built serverless infrastructure on AWS with SST and automated testing and deployment pipelines in GitLab
  - Worked closely with the non-technical project manager to ensure the project vision was met

  Tech: Next.js, TypeScript, Python, Drizzle, tRPC, TanStack Query, better-auth, Tailwind CSS, Docker, AWS, SST, SQL, GenAI

Developer — 09/2022 - 12/2024
inpro — Berlin, Germany

- **Factory Chatbot :**
  - Built a shop-floor chatbot with Nuxt frontend and Python backend with RAG
  - Created a data ingestion pipeline to index and embed documents for retrieval
  - Developed an internal web app for testing and benchmarking the RAG system
  - Deployed infrastructure to AWS with CDK and set up CI/CD pipelines in GitLab

  Tech: Nuxt, TypeScript, Python, Langchain, FastAPI, Drizzle, tRPC, Tailwind CSS, Docker, AWS, CDK, SQL, GenAI, RAG
- **Text2Tech (BMBF funded) :**
  - Built the frontend for a technology monitoring demonstrator that uses deep learning to extract information from automotive industry texts
  - The system automatically identifies technologies, companies, and their relationships from German and English sources

  Tech: React, TypeScript, Docker, GenAI

Junior Developer — 08/2020 - 08/2022
inpro — Berlin, Germany

- **Virtual Process Chain :**
  - Developed a C++ desktop application for data transfer between CAD/CAE systems
  - Created Python bindings for the library and integrated computer vision for scan mapping
  - Served as Scrum Master for the development team

  Tech: C++, Qt, cmake, Python, Mesh & Scan Data, Scrum Master
- **hypro (BMBF funded) :**
  - Built a Cadmould data input module for importing injection molding simulation data
  - Developed data mapping to transfer fiber orientation information from Cadmould simulations to Abaqus mesh
  - Created result evaluation in Python
  - Attended project meetings to present progress and findings

  Tech: C++, Python, Qt, cmake, Scrum Master

PROJECTS

- **gemhog.com :**
  - We listen to financial podcasts so you don't have to.
  - Investment ideas, trends, and expert takes — delivered to your inbox.
  - Pairs podcast theses with accessible financial data for evaluation

  Tech: Next.js, TypeScript, Effect-TS, SST, Postgres, AWS
- **Real Estate Calculator iOS App :**
  - Building an iOS app that simplifies real estate calculation for investment
  - Development stalled

  Tech: Expo, TypeScript, React Native, Hono, Drizzle, tRPC, TanStack Query, SST, Nativewind

CERTIFICATIONS

AWS Certified Machine Learning Engineer - Associate *AWS*	*2025*
Git Advanced Topics *GROSSWEBER*	*2023*
Clean Code C++ *GROSSWEBER*	*2022*
Improving Deep Neural Networks *Coursera*	*2020*
Machine Learning *Coursera*	*2020*
AWS Certified Solutions Architect - Associate *AWS*	*2025*
Professional Scrum Master I *Scrum.org*	*2023*
Design Thinking Practitioner *launchlabs Berlin*	*2021*
Neural Networks and Deep Learning *Coursera*	*2020*

EDUCATION

University of Applied Science Münster — 2016 - 2020
M.Sc. Chemical Engineering

- **Master's Thesis Student :**
  - At University of Berlin, I developed and implemented a viscoelastic fluid model in OpenFOAM's C++ codebase to simulate free-rising bubbles
  - Modified the bubbleInterTrackFoam solver by implementing an extra stress tensor and adapting the viscoelasticModel library
  - Performed pseudo-2D and 3D simulations to validate the implementation and optimize numerical parameters
  - I created visualizations and graphs with Python
- **Research Trainee :** 6-month traineeship at Universidad de Cadiz
- **Research Assistant :** At the University of Applied Science Münster

University of Applied Science Münster — 2012 - 2015
B.Sc. Chemical Engineering